

A multicriteria outranking approach for ship collision risk assessment

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ABSTRACT

The paper presents a judgmental expert-based process to assess the ship collision risk using the ELECTRE Tri-nC multiple criteria outranking method. The approach first establishes a set of criteria deemed important by navigation experts, all of them experienced nautical officers, to assess the collision risk between two ships in open waters. A deck of cards-based method is then used to determine the weights of each criterion, according to the experts' preferences. Finally, risk categories chosen by the experts are characterised in terms of multiple reference scenarios, also provided by the experts, and the ELECTRE Tri-nC method is used to assign validation scenarios to the risk categories. The results are compared with the experts' assessments of the collision risk for the same scenarios and also with the individual assessments made by three experts that have not participated in the development of the decision model.

1. Introduction

Maritime transportation involves risks such as fire, grounding, collision, loss of property and spillage of dangerous or pollutant substances [1,2]. Aside from the risk to life and property such as ships and their cargoes, there are also potentially disastrous consequences for the environment and for economic activities dependent on the sea, such as fisheries and tourism.

The risk management of maritime transportation is a very broad research area involving many disciplines that are typically addressed by different research teams focused on specific aspects [2]. Among the different disciplines, the collision risk has been widely addressed as typically ship collisions account for a large percentage of the total number of marine casualties [3]. According to the European Maritime Safety Agency (EMSA), ship collisions accounted for 26.2% of total marine casualties with ships between 2011 and 2018 [4].

The problem of assessing ship-to-ship collision risk has been addressed by numerous researchers. Different ship collision scenarios have been addressed in the literature, ranging from generic collisions between ships involved in traditional shipping activities (e.g. [5]) to scenarios involving ships operating near offshore wind farms (e.g. [6–8]), such as collisions between service vessels and offshore wind turbines [9], and also between shuttle tankers and oil and gas offshore fixed and floating platforms [10] and, more recently, scenarios

involving marine autonomous surface ships (e.g. [11]). Moreover, collision risk has been dealt with from different perspectives. Several studies address the collision risk using statistical approaches from historical data, which allows to characterize the main contributors to collisions and the development of risk models that represent with a probabilistic framework the influence of relevant risk influencing factors on collisions events [12–16] and on the individual risk of ships [17].

The introduction of the Automatic Identification System (AIS) has enabled the collection of large amounts of data on position, speed and course of the ships that have been used in several studies on marine traffic and associated risks [18], particularly for developing ship-to-ship collision risk models (e.g. [5]), as reviewed by Chen et al. [3]. As the occurrence of maritime accidents is relatively rare, the concepts of collision candidates (e.g. [19,20]) and near collisions or other non-accidental events (e.g. [21–23]), have been applied to assess the collision risk and its spatial distribution [24], which is typically associated with local maritime traffic characteristics, as shown by Rong et al. [25].

Determining collision risk is an essential part of the decision-making process that is necessary to ensure the safety of navigation. It is important both for officers responsible for the ship's navigation and for Vessel Traffic Service (VTS) operators, responsible for monitoring traffic in ports, inland waterways, or coastal areas. The International Regulations for Preventing Collisions at Sea (COLREGs) contain rules

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applicable to all vessels upon the high seas and in all waters connected therewith navigable by seagoing vessels. Among those rules, Rule 7 deals with the risk of collision. However, the rule provides little guidance on how to determine it, mentioning “such risk shall be deemed to exist if the compass bearing of an approaching vessel does not appreciably change”. According to Cockcroft and Lameijer [26], “The 1972 Conference rejected a proposed definition that risk of collision exists between vessels when their projected courses and speeds place them at or near the same location simultaneously”. These authors add that “in the Courts of the United Kingdom and other countries risk of collision has not been held to apply at long distances when there is a low speed of approach”.

The COLREGs may be considered subjective, in the sense that some scenarios may be interpreted differently and may produce different collision avoidance actions by navigators following the same set of rules [27]. Even the applicability or the moment when the Collision Regulations begin to apply are subjective and have been motive for interesting discussions. Zhao [28] states that “when the obligation of a privileged vessel to hold her course begins to attach or when meeting vessels should alter course depends not only on the distance between the vessels, but on the relation of their courses to each other, on their speed, at night on the range of visibility of their lights, on the character of the waters which they are navigating, and on other infinitely varying conditions”. In fact, these infinitely varying conditions result in an infinite number of possible scenarios. To further complicate matters, rule 2 (b) admits deviations from the Rules themselves, stating that “In construing and complying with these Rules due regard shall be had to all dangers of navigation and collision and to any special circumstances, including the limitations of the vessels involved, which may make a departure from these Rules necessary to avoid immediate danger”.

The lack of a definition for Risk of Collision in the COLREGs does not necessarily mean that there is not common understanding of its meaning among mariners. One can argue that the “ordinary practice of seamen” mentioned in the rules acknowledges the existence of that common understanding. However, how the assessment of the Risk of Collision is made may vary with the mariner’s training and experience. The reason why the Risk of Collision is not defined in the COLREGs may be related with the complexity of defining risk in general. The Society for Risk Analysis Glossary [29] acknowledges the existence of different definitions of concepts fundamental for the risk analysis field. This document includes seven different overall qualitative definitions of risk and six different examples of risk metrics/descriptions, stating “experience has shown that to agree on one unified set of definitions is not realistic”.

Kaplan and Garrick [30] state that risk is relative to the observer’s perception, which is subjective and depends on who is performing the assessment, and searching for an “absolute risk” only leads to somebody else’s perceived risk. Defining risk as relative to a perception falls into the constructivist risk paradigm, which, along with the realist paradigm, is one of the two higher-order paradigms for the concept of risk. Within the realist paradigm, risk is understood as a quantifiable characteristic of an event. In contrast, within the constructivist paradigm, risk is considered a construct that has its origins in the human mind, a perception with a subjective nature [31]. Constructivist approaches require expert elicitation to develop risk models, while realist approaches are more reliant on system data and/or engineering models [32,33].

Goerlandt and Montewka [32] have reviewed methods and applications for maritime transportation risk analysis in light of some foundational issues, characterizing applications according to their characteristics, including the analysis aim and scope, definition of risk, types of evidence and the adopted approach to risk analysis science, among others. Regarding the possible different approaches to risk analysis science, these authors considered eight different classes, ranging from strong realist to strong constructivist approaches.

When evaluating the collision risk, the Distance to Closest Point of Approach (DCPA) and Time to Closest Point of Approach (TCPA) parameters [24,34] are the most frequently used. Other collision criteria

have been proposed such as the Minimum Distance To Collision (MDTC) criterion by Montewka et al. [35], which uses a ship dynamics model to take into account ships’ manoeuvrability characteristics when calculating a critical distance under which collision avoidance actions cannot prevent the collision from occurring, and a Time Discrete Non-linear Velocity Obstacle method by Chen et al. [36], in which a collision candidate is defined as an encountering pair of ships such that the velocity obstacle is violated by the own ship during a certain period, rather than resulting from spatio-temporal relation between ships at certain time instances. Several approaches have been adopted to define collision risk, which is used as the initiation of automatic collision avoidance actions in different systems [37–40].

However, there is no common ground to define collision risk, due to the complexity of the marine environment, lack of information and the large number of feature parameters [41]. Different mental models of risk, affected by the different training, experience and interpretation of the rules, result in different assessments of the risk of collision. In this paper, a constructivist approach is adopted that addresses the ship collision risk assessment as a decision problem, where encounter scenarios are assigned to risk categories. The ELECTRE Tri-nC multicriteria outranking method [42] is adopted to assess ship collision risk based on a judgmental expert-based process.

2. Ship collision risk assessment problem

As previously referred, although the risk of collision is explicitly mentioned in the COLREGs, no definition is provided. However, even without a legal definition or clear guidelines on how to assess collision risk, mariners repeatedly perform that assessment and take actions to keep collision and other risks within acceptable levels. Goerlandt et al. [33] presented an overview of measures used in methods for collision risk assessment and in Collision Alert Systems (CAS), which includes: measures of how ships are positioned and moving relatively to each other, such as Distance at Closest Point of Approach (DCPA), Time to Closest Point of Approach (TCPA), relative bearing, relative heading, range, variation of bearing, ship speed and COLREG status; measures of ships’ characteristics, such as ship length and gross tonnage (GT); and measures of environmental conditions, such as waves, visibility, day/night time and operating area. Any vessel required to comply with the International Convention for the Safety of Life At Sea (SOLAS) must be equipped with a radar system which must be capable of tracking targets and include a CAS based on distance at CPA and TCPA. This system must also be able to display information provided by the Automatic Identification System (AIS). Radar and AIS systems provide real-time information on positions, courses and speeds, and on predicted CPAs and TCPAs, which can be displayed in a graphical interface, adjustable to the navigator’s preferences.

However, assessing collision risk is not just a kinematics problem. If the perceived risk in a given situation can be related to the number and quality of the options available to reduce risk, then other factors can play a significant role in the risk assessment process, such as: ship manoeuvrability, which can be related to characteristics such as dimensions, type and loading condition, among others; severe weather conditions, which can prevent a ship from manoeuvring or from navigating at certain headings and hinder visual and radar detection; the COLREGs and how they are learned, understood and interpreted by those who use the sea for activities so diverse as carrying cargo, transporting people, fishing or leisure; restrictions such as land masses, shoals, obstacles, channels or routing measures; or the way a potential action can affect the collision risk with other vessels.

Among all factors that can be relevant when assessing collision risk, the COLREGs and how they are interpreted may be the largest source of uncertainty. Belcher [43] used a situation involving three vessels to call into question the certainty and predictability of collision avoidance under the COLREGs. According to the author, because each ship must independently choose mutually compatible strategies, it is hardly

surprising that many collisions at sea occur after one or both vessels take action to avoid collision, adding that the fact that each ship has to understand the actual, likely and potential actions of the other lies at the heart of the problem of applying the COLREGs. To add even more complexity, the assumption that everyone is following the same set of rules may not hold, because training and certification requirements to operate small fishing or recreational crafts are not universal. Belcher [43] concludes that prescriptive rules are insufficient as a means of risk management and can never cover all situations as the variety of rules developed to govern safe behaviour will always be less than the possible variety of unsafe conditions, stating that safety at sea would be improved by removing the rules that cannot be adequately used in practice, and not through the addition of rules.

Knowledge and interpretation of the COLREGs can be considered a human factor in collision risk assessment and collision avoidance. Human and organizational factors in ship collision risk assessment include: personal factors such as cognitive fatigue or stress; aspects of communication information such as lack of information; the influence of internal/external environmental factors such as weather or the time of day; organizational factors such as manning characteristics or organizational policies; training/competence issues such as no training or lack of experience [13].

Real time collision risk assessment is not an exclusive activity to those operating ships. Vessel Traffic Services (VTS) contribute to the safety of life at sea, the safety and efficiency of navigation and protection of the marine environment, adjacent shore areas, work sites and offshore installations from possible adverse effects of maritime traffic, by providing three types of service: Information Service; Traffic Organization Service; and Navigational Assistance Service. VTS operators are often required to assess collision risk in order to provide information or assistance, depending on the type of service provided and on circumstances. A different type of collision risk assessment, not in real time, may be of interest to decision makers when considering the introduction or changes in traffic organization or routeing measures, or when deciding the location of Search And Rescue (SAR) or pollution response assets.

Goerlandt et al. [33] identified three underlying mechanisms in interpreting risk: imminence to accident occurrence; deviation from a reference level; and ambiguous situations. Since in the final stage of any collision between two ships, these must be in a collision course with no possibility of performing a collision avoidance action [35], the concept of collision candidate, an encounter which will result in a collision if nothing is done to prevent it, is often used to assess the imminence of a collision [19,35,44,45]. An example of a deviation from a reference level can be found in violations of the ship safety domain, a concept defined as an area around a vessel that the navigator wants to keep free of other vessels [46–48]. As for ambiguous situations, these mostly result from different interpretations of an encounter situation in light of the COLREGs, as analysed by Belcher [43]. All three mechanisms described were mentioned by the experts during the development of the decision model presented in this paper. Furthermore, the experts' mental model itself can be considered a reference level, to which encounter scenarios are compared to assess collision risk.

3. An overview of the ELECTRE TRI-NC method

The opinion of experts is a valued source of information in the maritime domain (e.g. [49]), and in collision risk analysis in particular. Assessing ship collision risk can be approached as a decision problem, where experts take the place of decision makers, who must assign a risk class to a given ship encounter scenario.

Multiple Criteria Decision Aiding (MCDA) methods have successfully been used as decision support tools in various fields for decades. In particular, the ELECTRE (ELimination Et Choix Traduisant la REalité – Elimination and Choice Expressing the Reality) family of methods, developed in the 1960's, have been applied in a wide range of problems

due to their distinctive features. These methods use as a preference model an outranking relation on the set of actions and include strong features such as the possibility of dealing with the qualitative as well as the quantitative scales of criteria, the heterogeneity of scales, the non-relevance of compensatory effects and can take into account the imperfect knowledge of data and some arbitrariness when building criteria [50]. A detailed discussion of the weak and the strong points of ELECTRE methods can be found in [51].

Among the ELECTRE methods, the ELECTRE Tri-NC [42] is of particular interest as it allows for various reference actions per category, and is therefore better suited to model collision risk decision problems by assigning several reference encounter scenarios (actions) to each risk category.

3.1. ELECTRE methods

A decision aiding process results from the interaction of two key actors: the decision-maker and the analyst, who provides the decision aiding. In this context, the decision aiding activity is founded on three pillars [50]:

- The *actions* that are the objects of the decision;
- The *consequences*, which are the attributes of the actions, allowing the comparison of different actions;
- The *modelling of one or several preference systems*, consisting of an implicit or explicit process that assigns to each pair of actions, one and only one of the three situations: *indifference*, *preference*, or *incomparability*.

Any MCDA problem needs the following input data [50]:

- A set of potential actions $A = \{a_1, \dots, a_i, \dots, a_m\}$;
- A coherent family of criteria $F = \{g_1, \dots, g_j, \dots, g_n\}$;
- The performance of each potential action a_i on each criterion g_j , denoted as $g_j(a_i)$.

The relative importance of each criterion g_j , also known as *weight*, is denoted as w_j and can be interpreted as the voting power of each criterion. ELECTRE methods assume four possible outcomes when comparing two actions:

- Indifference*, when there is an equivalence between two actions;
- Strict preference*, when one action is clearly favoured over the other;
- Weak preference*, when one action is not clearly favoured over the other, but they are not equivalent (in other words it is a situations where there is "hesitation" between strict preference and indifference);
- Incomparability*, when none of the previous situations is possible.

In ELECTRE methods, criteria are considered pseudo-criteria to take into account the imperfect character of data. A pseudo-criterion is defined by a g_j associated to two thresholds (we shall consider the version with constant thresholds): an *indifference threshold*, q_j , and a *preference threshold*, p_j , such that $p_j \geq q_j \geq 0$ [42].

For any ordered pair of actions $(a, a') \in A \times A$, binary relations are defined for any pseudo-criterion $g_j \in F$:

$$g_j(a) - g_j(a') > p_j(g_j(a')) \Leftrightarrow aP_ja' \quad (1)$$

$$q_j(g_j(a')) < g_j(a) - g_j(a') \leq p_j(g_j(a')) \Leftrightarrow aQ_ja' \quad (2)$$

$$-q_j(g_j(a)) \leq g_j(a) - g_j(a') \leq q_j(g_j(a')) \Leftrightarrow aI_ja' \quad (3)$$

A partial outranking relation $S_j = P_j \cup Q_j \cup I_j$ on criterion g_j results from the grouping of these three binary relations, where aS_ja' means that "a is at least as good as a'" on criterion g_j .

A comprehensive outranking relation S , meaning “at least as good as”, models preference in ELECTRE methods, with four possible outcomes:

- aSa' and $not(a'Sa)$, meaning a is weakly or strictly preferred to a' ;
- $a'Sa$ and $not(aSa')$, meaning a' is weakly or strictly preferred to a ;
- aSa' and $a'Sa$, meaning a is indifferent to a' , aIa' ;
- $not(aSa')$ and $not(a'Sa)$, meaning a is incomparable to a' , aRa' .

Each criterion (or pseudo-criterion) g_j has a relative importance coefficient w_j , commonly referred as weight. This intrinsic weight can be interpreted as the voting power of a criterion. It is assumed that $\sum_{j=1}^n w_j = 1$. A criterion's importance grows as w_j increases [50].

The concepts of *concordance* and *discordance* are used in outranking based methods. Let $C(aIa')$ be the subset of criteria such that aI_ja' , $C(aPa')$ be the subset of criteria such that aP_ja' and $C(aQa')$ be the subset of criteria such that aQ_ja' . The comprehensive concordance index, representing the strength of the coalition of criteria being in favour of aSa' , is defined as:

$$c(a, a') = \sum_{j \in C(aPa')} w_j + \sum_{j \in C(aQa')} w_j + \sum_{j \in C(aIa')} w_j + \sum_{j \in C(a'a)} w_j \varphi_j \quad (4)$$

where

$$\varphi_j = \frac{g_j(a) - g_j(a') + p_j(g_j(a))}{p_j(g_j(a)) - q_j(g_j(a))} \in [0, 1] \quad (5)$$

The partial discordance indices $d_j(a, a')$, $j = 1, \dots, n$, that may take into account veto thresholds v_j , such that $v_j \geq p_j$, are defined as:

$$\begin{cases} 1 & \text{if } g_j(a) - g_j(a') < -v_j(g_j(a)) \\ \frac{g_j(a) - g_j(a') + p_j(g_j(a))}{p_j(g_j(a)) - v_j(g_j(a))} & \text{if } -v_j(g_j(a)) \leq g_j(a) - g_j(a') < -p_j(g_j(a)) \\ 0 & \text{if } g_j(a) - g_j(a') \geq -p_j(g_j(a)) \end{cases} \quad (6)$$

The comprehensive outranking of a over a' is denoted by the credibility index $\sigma(a, a')$, which reflects the strength of aSa' , and is defined as:

$$\sigma(a, a') = c(a, a') \prod_{j=1}^n T_j(a, a') \quad (7)$$

where

$$T_j(a, a') = \begin{cases} \frac{1 - d_j(a, a')}{1 - c(a, a')} & \text{if } d_j(a, a') > c(a, a') \\ 1 & \text{otherwise} \end{cases} \quad (8)$$

Note that the indifference, preference and veto thresholds may vary with the performances $g_j(a)$ or $g_j(a')$ [52].

3.2. ELECTRE Tri-nC method

The ELECTRE Tri-nC is a sorting method that takes into account several reference actions for characterizing each category [42]. It builds on the assumptions that there is an ordered set of categories $C = \{C_1, \dots, C_h, \dots, C_q\}$, prepared to receive actions, where each category is characterized by a subset of reference actions $B_h = \{b_h^r, r = 1, \dots, m_h\}$, which represent the actions that should be assigned to that category, according to the decision maker. Two additional subsets are necessary to apply the method, containing one reference action each: $B_0 = \{b_0^1\}$ and $B_{q+1} = \{b_{q+1}^1\}$. These reference actions are defined as follows: $g_j(b_0^1)$ is the worst possible performance on criterion g_j , and $g_j(b_{q+1}^1)$ is the best possible performance on the same criterion g_j , for all $g_j \in F$ [42].

Since an action must be compared to sets of reference actions to be assigned to a category, it is necessary to define categorical credibility indices:

$$\sigma(\{a\}, B_h) = \max_{r=1, \dots, m_h} \{\sigma(a, b_h^r)\} \quad (9)$$

$$\sigma(B_h, \{a\}) = \max_{s=1, \dots, m_h} \{\sigma(b_h^s, a)\} \quad (10)$$

A minimum credibility level, denoted by λ , may be considered necessary to validate the statement “ a outranks B_h ” when all criteria are taken into account. This level takes a value within $[0.5, 1]$, and is used to define four possible outcomes when comparing an action with the reference actions in a category, defined as λ -binary relations:

- λ -outranking: $\{a\}S^\lambda B_h \Leftrightarrow \sigma(\{a\}, B_h) \geq \lambda$;
- λ -preference: $\{a\}P^\lambda B_h \Leftrightarrow \sigma(\{a\}, B_h) \geq \lambda \wedge \sigma(B_h, \{a\}) < \lambda$;
- λ -indifference: $\{a\}I^\lambda B_h \Leftrightarrow \sigma(\{a\}, B_h) \geq \lambda \wedge \sigma(B_h, \{a\}) \geq \lambda$;
- λ -incomparability: $\{a\}R^\lambda B_h \Leftrightarrow \sigma(\{a\}, B_h) < \lambda \wedge \sigma(B_h, \{a\}) < \lambda$.

Categories must fulfil at least a dominance condition and a weak separability condition, as defined by Almeida-Dias et al., [42].

The procedure to assign an action to a category is composed of an ascending rule and a descending rule, which must be used jointly. A selecting function ρ is defined as:

$$\rho(\{a\}, B_h) = \min\{\sigma(\{a\}, B_h), \sigma(B_h, \{a\})\} \quad (11)$$

The descending rule is defined as follows: decrease h from $(q+1)$ to t , which is the first value such that $\sigma(\{a\}, B_t) \geq \lambda$, making C_t the descending pre-selected category. For $t = q$, select C_q as a possible category to assign action a to. For $0 < t < q$, if $\rho(\{a\}, B_t) > \rho(\{a\}, B_{t+1})$, select C_t as a possible category to assign a to; otherwise, select C_{t+1} . For $t = 0$, select C_1 as a possible category to assign a to.

The ascending rule is defined as follows: increase h from zero to k , which is the first value such that $\sigma(B_k, \{a\}) \geq \lambda$, making C_k the ascending pre-selected category. For $k = 1$, select C_1 as a possible category to assign action a to. For $1 < k < (q+1)$, if $\rho(\{a\}, B_k) > \rho(\{a\}, B_{k-1})$, select C_k as a possible category to assign a to; otherwise, select C_{k-1} . For $k = (q+1)$, select C_q as a possible category to assign a to. An action a can be assigned to:

- one category, when both rules select the same category;
- two categories, when the two rules select consecutive categories;
- a range of more than two consecutive categories, delimited by the categories selected by the two rules.

4. Expert based multiple criteria outranking approach to collision risk assessment

4.1. Defining the risk classification problem with Experts

A panel of five experts acted as decision makers in the collision risk model building process, all of them experienced merchant marine officers, having previously served as Masters or mates on cargo ships, tankers, cruise ships, fishing vessels, car carriers, dredgers, offshore support vessels and tugs. All experts have received education and training to become merchant marine officers according to the Standards of Training, Certification and Watchkeeping (STCW) Code, and are themselves training future officers at Escola Superior Náutica Infante D. Henrique (ENIDH), which is the higher education school for maritime studies in Portugal.

In order to clarify the meaning of ship collision risk in the scope of this work, and to establish a common ground between all experts and the decision-aiding analysts, the interpretation of COLREGs, in particular Rule 7, by an experienced deck officer is adopted by all participants to define the “risk of collision”.

The problem is restricted to encounters between two ships with Gross

Tonnage (GT) equal or greater than 500 in open sea, with no land, depth restrictions or obstacles in the vicinity, and with the distance between both ships decreasing. These can be considered ideal scenarios, adequate for developing a first iteration of the risk assessment model, which is therefore applicable to any waters fitting the described conditions. The risk classification of an encounter scenario is to be made from the perspective of the officer in charge of navigation of a ship denoted as Own Ship (OS), while observing a Target Ship (TS).

4.2. Criteria modelling

The experts were asked what criteria would be relevant when assessing and classifying the risk of collision, taking into account the aforementioned context and restrictions. The discussion that followed resulted in a preliminary list of criteria, in no particular order:

- **Closest Point of Approach (CPA):** also referred in some literature as Distance at CPA (DCPA). This is the predicted minimum distance between the two vessels if there are no changes in course or speed;
- **Time to Closest Point of Approach (TCPA):** the remaining time until the CPA is reached;
- **OS type:** the Own Ship type, e.g. tanker, passenger ship, fishing vessel, etc.;
- **TS type:** the Target Ship type;
- **Wind and sea conditions:** affect the visual and radar detection capability as well as the ship manoeuvrability. In severe weather, it may not be possible to execute certain course or speed changes to avoid a collision or to increase the CPA;
- **Visibility:** distance at which a Target Ship can be sighted from the Own Ship. It is affected by fog, rain, sea spray, among other factors;
- **OS dimensions and displacement:** the dimensions and displacement of a ship affect the time and distance needed to take actions to avoid a collision or to increase CPA. In general, larger ships need more time to alter course and speed;
- **TS dimensions and displacement;**
- **OS navigational status:** depends on the operations being performed by the ship, which may limit the ship's capability to alter course or speed;
- **TS navigational status:** depends on the operations being performed by the ship, which may limit the ship's capability to alter course or speed or affect the predictability of a Target Ship's actions;
- **Duty to take action:** whether or not the OS is required to take action to avoid a collision or a close quarters situation, according to the COLREGs. This is affected by their relative positions, headings, speeds and navigational status as well as by the visibility. In some cases, a ship that is initially not required to take action to avoid a collision must do so if there is no action taken by the other vessel, or the action taken by the other vessel is insufficient. The moment when this change in behaviour should occur is not clearly stated in the rules and is one of many subjective aspects of the COLREGs;
- **Daylight:** during the night it may be harder to visually determine a Target ship's heading, type, dimensions and navigational status. To do so, mariners rely on navigational lights, which must be shown in compliance with the COLREGs;
- **Relative bearing:** the bearing of the Target ship, relative to the Own Ship's heading;
- **Distance:** distance between OS and TS;
- **Bow crossing:** whether the direction of the TS motion relative to the OS crosses the bow or the stern of the OS.

The next task is to determine how to assess the performance of each criterion. Some of the criteria are merged during the process. The revised list of criteria and their performance evaluation methods are described in the next sub-sections, as well as the procedure used to determine the criteria weights.

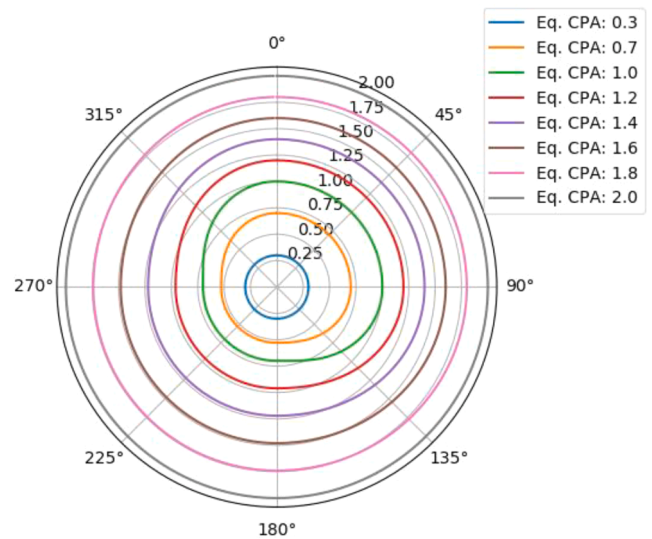


Fig. 1. Equivalent CPA.

4.2.1. Equivalent CPA

This criterion resulted from merging the CPA and Bow Crossing preliminary criteria, while incorporating some aspects of the Duty to Take Action preliminary criterion. The experts all agreed that a scenario where the direction of relative motion crosses the bow of the OS contributed to a higher risk perception than a scenario where the direction of relative motion crosses astern of the OS. However, since ships on parallel courses or on a collision course do not cross each other's bow or stern, a performance measurement for the Bow Crossing criteria could not be established. Instead, the experts decided to take advantage of the fact that the CPA is located ahead of the beam of the OS when the direction of relative motion crosses the bow of the OS and astern of the beam when the direction of relative motion crosses the stern of the OS. When the two vessels are on parallel courses, the CPA is located on the beam of the OS.

The experts were asked what values of CPA would make the CPA position irrelevant, either because the risk is high enough, or because the risk is low enough. The experts stated that for CPAs equal or less than 0.3 nautical miles (nm) and for CPAs equal or greater than 2 nautical miles, the position where the CPA would occur would be irrelevant, because the risk would be too high or too low, respectively, to make a difference. It is reasonable to assume that these standards may vary according with the type of waters and traffic conditions. However, in this particular case, as with all the following criteria, the near ideal open waters conditions established when defining the problem apply. Taking these limits into account, some parallel course scenario comparisons were conducted, with the OS overtaking the TS and being overtaken by the TS, both on port side and on starboard side. Furthermore, opposite courses scenario comparisons were conducted, where the TS would pass both on port side and on starboard side of the OS.

The predicted CPAs for these scenarios were close to 1 nautical mile. The experts concluded that a CPA of 1 nautical mile on the starboard beam of the OS is equivalent to a CPA of 0.7 nautical miles on the port beam of the OS, due to the fact that the starboard side would remain clear in case a change in course was necessary. Another set of comparisons were conducted between scenarios where the direction of relative motion is perpendicular to the OS's heading, crossing either ahead or astern of the OS, with CPAs around 1 nautical mile. The experts concluded that a CPA of 1 nautical mile on the bow of the OS is equivalent to a CPA of 0.7 nautical miles directly astern of the OS, because in the first case the OS is moving towards the CPA position and in the second case the OS is moving away from the CPA position.

Comparisons between parallel course scenarios and scenarios where

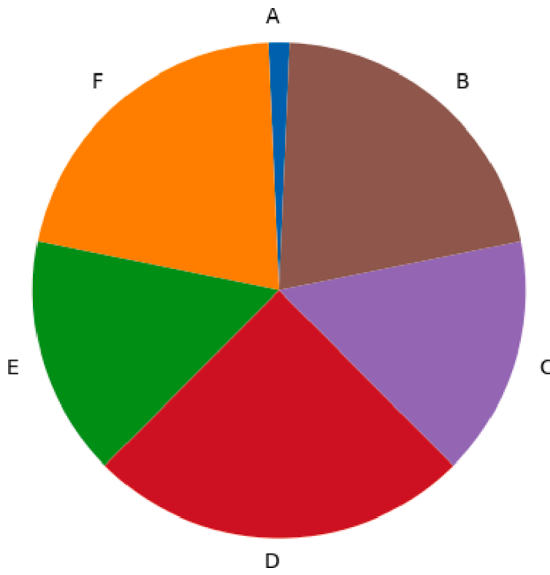


Fig. 2. Relative bearing sectors.

the direction of relative motion is perpendicular to the OS's heading resulted in the conclusion that CPAs of 1 nautical mile on the bow, 1 nautical mile on the starboard beam, 0.7 nautical miles on the stern and 0.7 nautical miles on the port beam are equivalent. The following equivalent CPA function, based on the experts' opinions, was proposed and accepted. The graphical representation of the Equivalent CPA function for increasing CPA values ranging from 0.3 to 2 nautical miles, is shown in Figure 1.

$$CPA_{eq} = CPA \text{ if } CPA \leq 0.3 \text{ nm} \quad (12)$$

$$CPA_{eq} = 0.3 + \frac{0.7(CPA - 0.3)}{(dz - 0.3)} \text{ if } 0.3 \text{ nm} < CPA \leq dz \quad (13)$$

$$CPA_{eq} = 1 + \frac{(CPA - dz)}{(2 - dz)} \text{ if } dz < CPA < 2 \text{ nm} \quad (14)$$

$$CPA_{eq} = CPA \text{ if } CPA \geq 2 \text{ nm} \quad (15)$$

with:

$$dz = \sqrt{\sin(Z_{CPA})^2 + \cos(Z_{CPA})^2} \text{ if } 0 \leq Z_{CPA} < \frac{\pi}{2} \quad (16)$$

$$dz = \sqrt{\sin(Z_{CPA})^2 + (0.7\cos(Z_{CPA}))^2} \text{ if } \frac{\pi}{2} \leq Z_{CPA} < \pi \quad (17)$$

$$dz = \sqrt{(0.7\sin(Z_{CPA}))^2 + (0.7\cos(Z_{CPA}))^2} \text{ if } \pi \leq Z_{CPA} < \frac{3\pi}{2} \quad (18)$$

$$dz = \sqrt{(0.7\sin(Z_{CPA}))^2 + \cos(Z_{CPA})^2} \text{ if } \frac{3\pi}{2} \leq Z_{CPA} < 2\pi \quad (19)$$

where Z_{CPA} is the bearing of the CPA relative to the OS heading.

4.2.2. TCPA

This is the remaining time until the predicted CPA occurs, usually showed in minutes on the ship's Automatic Radar Plotting Aid (ARPA) and other navigation equipment. The experts indicated that the risk of collision increases as the TCPA decreases.

4.2.3. Distance

The distance between the OS and a TS is commonly measured in nautical miles. One nautical mile is defined as 1852 metres. The experts stated that the risk of collision increases as the distance decreases.

4.2.4. Relative bearing

Some aspects regarding the duty to take action according to the COLREGs, which was initially mentioned as a possible criterion by the experts, are considered when defining a performance evaluation method for the Relative Bearing. Relative bearings have the heading of the OS as reference and are commonly measured in degrees. The experts decided to define six sectors with six different risk levels, defined as follows (Fig. 2):

- Sector A: 5° - $[357.5^\circ, 002.5^\circ]$;
- Sector B: 76° - $[002.5^\circ, 078.5^\circ]$;
- Sector C: 56.5° - $[078.5^\circ, 135^\circ]$;
- Sector D: 90° - $[135^\circ, 225^\circ]$;
- Sector E: 56.5° - $[225^\circ, 281.5^\circ]$;
- Sector F: 76° - $[281.5^\circ, 357.5^\circ]$

Regarding the risk of collision, the experts' judgement on the order of the sectors, from higher risk to lower risk, is C, E, F, B, A, D.

4.2.5. Environmental conditions

The wind and sea conditions, as well as the visibility and daylight conditions were all identified as preliminary criteria by the experts. Mariners usually measure the wind and sea conditions using the Beaufort wind force scale, which relates wind speed to observed sea conditions. The Beaufort scale ranges from force zero (calm) to force twelve (hurricane force). Wind and seas affect a ship's speed, the ability to alter course and the visual and radar detection of target vessels.

Visibility at sea is commonly measured in nautical miles. It can be affected by fog, haze, rain and sea spray, among other factors. The COLREGs include rules that apply to ships in sight of one another, a rule for the conduct of vessels in restricted visibility and rules that apply in any condition of visibility. When two vessels cannot sight each other, they can use radar, ARPA, Automatic Identification System (AIS) or any other means to detect each other and to support a possible decision to alter course or speed to avoid a collision or to increase the predicted CPA. Nevertheless, the experts considered that sighting a target ship in an encounter situation decreases the risk of collision.

During daytime, when two vessels can sight each other, mariners can directly observe a clear course or speed change in a target ship. At night, ships use lights of different colours, visible from different sectors, which give information to other vessels about their heading and other relevant features. The experts' opinion was that the absence of daylight increases the risk of collision.

The experts decided to combine the wind and sea state, the visibility and the daylight condition into one criterion, named environmental conditions. After discussing the effects of the three factors, the experts agreed on eight possible classifications for this criterion, ranked from higher to lower risk as follows:

- E8: Beaufort force 9 or more, night-time;
- E7: Beaufort force 9 or more, daytime;
- E6: Beaufort force 7 or 8, target not visible;
- E5: Beaufort force 6 or less, target not visible;
- E4: Beaufort force 7 or 8, night-time, target visible;
- E3: Beaufort force 7 or 8, daytime, target visible;
- E2: Beaufort force 6 or less, night-time, target visible;
- E1: Beaufort force 6 or less, daytime, target visible;

It was stated by the experts that for Beaufort force 9 or more the visibility is always affected. For Beaufort force 8 or less, it was stated that when the target is not visible, the daylight condition is irrelevant.

4.2.6. Own ship length

The experts decided to use the length as an indicator of the OS's dimensions and displacement. A ship's length can be measured in meters, and the experts considered that an increase in length increases the

Table 1
Normalized weights.

Criterion	Rank	Blank cards	Normalized weight
Equivalent CPA	1		21.5
TCPA	1		21.5
Distance	1		21.5
		3	
Relative bearing	2		14.7
		2	
Environmental conditions	3		9.6
		2	
OS length	4		4.5
OS status	4		4.5
		1	
TS length	5		1.1
TS status	5		1.1

time and distance needed to take action in an encounter situation between two vessels, having an increasing effect on the risk of collision.

4.2.7. Own ship status

It was decided to restrict the OS's status to three classes, ranked from higher to lower risk as follows:

- R: Vessel restricted in her ability to manoeuvre, which is a ship that, due to the nature of her operations, is unable to keep out of the way of another vessel;
- F: Vessel engaged in fishing, which is a vessel with fishing apparatus that restrict manoeuvrability;
- U: Vessel underway using engine, which is a power-driven vessel that is not at anchor, or made fast to shore or aground.

4.2.8. Target ship length

An increase in the target ship length, measured in metres, has a similar effect on the TS as described for the OS's length. From the OS's perspective, a larger TS is more difficult to avoid in a close quarters situation. For these reasons, an increase in TS length has an increasing effect on the risk of collision.

4.2.9. Target ship status

Four classes were defined to measure the target ship status, ranked from higher to lower risk as follows:

- F: Vessel engaged in fishing;
- R: Vessel restricted in her ability to manoeuvre;
- S: Sailing vessel, which is a vessel under sail, not using propelling machinery, if fitted;
- U: Vessel underway using engine

4.2.10. Criteria weights

The revised Simos' procedure developed by Figueira and Roy [53] is used to aid the experts in determining the criteria weights. This procedure uses a deck of cards, which contains one card per criterion with the criterion name written on the card, plus a sufficient number of blank cards, with no names written on them. The experts are asked to rank the criteria, from most important to least important, using the criteria cards as a visual aid. The difference of importance between the obtained ranks is expressed by introducing blank cards between the criteria cards assigned to each rank. The experts are asked how many times the most important criterion (or criteria) is more important than the least important criterion (or criteria), as this difference, denoted as Z , is necessary to apply the chosen method.

The non-normalized and the normalized weights are then calculated. In this particular application, the obtained normalized weights are shown in Table 1, with one decimal place.

Table 2
Quantitative criteria performance limits.

Criterion	Min. limit	Max. limit
Eq. CPA	0 nm	3 nm
TCPA	0 min.	60 min.
Distance	0 nm	12 nm.
OS length	40 m	400 m
TS length	40 m	400 m

4.3. Risk categories and reference scenarios

During the initial discussion with the experts five risk categories were proposed. However, the experts decided that, from a practical point of view, three risk categories or levels are more useful and would be in line with other risk classification systems used in the maritime domain such as the *Security Levels* of the International Ship and Port Facility Security (ISPS) Code or the *Ship Risk Profile* of the New Inspection Regime of the Paris Memorandum of Understanding on Port State Control that is used for ship selection in Port State Control inspections.

Thus, three risk categories are adopted: High Risk (HR), Medium Risk (MR) and Low Risk (LR). These categories are ready to receive actions, which in this application are scenarios involving two vessels: the OS and the TS. To characterize each category, reference scenarios must be defined. To aid in this process, experts were asked for minimum and maximum limits for the quantitative criteria (Equivalent CPA, TCPA, Distance, OS length and TS length). Those limits are shown in Table 2.

With the limits set, the experts were challenged to imagine, or recall from their experience, scenarios involving two ships that would fit into and correctly represent each of the three risk categories. The reference scenarios, which are the ELECTRE Tri-nC reference actions, are shown in Table 3. Note that on the High Risk category there is a worst case scenario (HR1) proposed by the experts, which has the "best" (highest risk) performance possible on every criteria except the TS length and the OS length, which have their "best" performances on scenarios HR2 and HR3, respectively. As for the Low Risk category, there is a scenario (LR3) with the "worst" (lowest risk) performance possible on all criteria. Nevertheless, it can be considered representative of the risk category, since scenarios with larger values of Eq. CPA, TCPA or distance, or with smaller values of OS and TS length are possible and would still be classified as Low Risk. Some of the scenarios proposed by the experts were rejected, in order to fulfil the dominance condition [42]. Scenarios HR1 to HR4 characterize the High-Risk category, scenarios MR1 to MR4 characterize the Medium Risk category and scenarios LR1 to LR3 characterize the Low Risk category.

4.4. Indifference, preference, and veto thresholds

In this section, the procedure used to elicit the indifference, preference and veto thresholds is described, and their respective values are presented.

4.4.1. Indifference and preference thresholds

Indifference and preference thresholds were defined by the experts only for the quantitative criteria. A criterion's indifference threshold is the maximum value of the performance difference of two actions for which the situation of indifference holds. A criterion's preference threshold is the smallest performance difference for which one action is preferred over the other [50].

For each of the quantitative criteria, the experts were asked to choose two different performance values (RP1 and RP2) that are clearly distinct from the best and worst possible performances. For each of the chosen performance values, the experts defined the indifference (q) and preference (p) thresholds.

These values are used to calculate the parameters α_q and β_p , which

Table 3

Reference actions or scenarios.

	Equivalent CPA (nm)	TCPA (min.)	Distance (nm)	Relative Bearing	Environ. Conditions	OS length (m)	OS status	TS length (m)	TS status.
HR1	0.1	5	1.13	F	E6	300	Underway	140	Fishing
HR2	0	0	0	C	E8	275	Restricted	400	Restricted
HR3	0.2	5	0.6	E	E6	400	Underway	300	Restricted
HR4	0.65	7	2.36	F	E5	275	Underway	200	Restricted
MR1	1	15	4.6	F	E2	275	Underway	140	Sailing
MR2	1.31	13	4.89	F	E4	250	Underway	140	Restricted
MR3	0.9	25	2.64	B	E4	260	Underway	125	Underway
MR4	0.7	8	4.05	B	E2	250	Underway	120	Underway
LR1	3	30	9.96	B	E1	150	Underway	115	Underway
LR2	2	60	5.6	D	E2	250	Underway	100	Underway
LR3	3	60	12	D	E1	40	Underway	40	Underway

Table 4

Indifference (q) and preference (p) thresholds.

	Eq. CPA	TCPA	Dist.	OS length	TS length
RP1	0.5	15	3	150	250
q	0.1	3	0.5	30	50
p	0.2	7	1	60	200
RP2	1.5	40	8	250	250
p	0.2	5	1	50	50
q	0.3	15	2	75	200
α_q	0.1	0.08	0.1	0.2	0
β_q	0.05	1.8	0.2	0	50
α_p	0.1	0.32	0.2	0.15	0
β_p	0.15	2.2	0.4	37.5	200

Table 5

Veto threshold (v) for criterion Equivalent CPA.

	Equivalent CPA
RP1	0.5
v	0.3
RP2	1.5
v	0.4
α_v	0.1
β_v	0.25

define the indifference threshold linear functions for each criterion, as well as the parameters α_p and β_p , which define the preference threshold linear functions for each criterion. The results are shown in Table 4.

Table 6

Experience and qualifications of external experts.

	Expert 1	Expert 2	Expert 3
STCW	Chief Mate	Chief Mate	Master <3000 GT
Certificate of Competence	>3000 GT	>3000 GT	
Years as mariner	10	4	21
Ship types	Passenger ships, tugs, PSV, general cargo, yachts	Container carriers, general cargo, HSC, ROPAX, yachts	Passenger ships, container carriers, bulk carriers, tugs, offshore DSV, sailing tall ship, yachts
Age	32	33	44

4.4.2. Veto threshold

The power of each criterion is defined by its weight and, when appropriate, by its veto threshold [52]. A criterion with veto power can prevent an action from being assigned to a given category [54]. In this application, the experts decided that the Equivalent CPA criterion should have this power because scenarios with a large enough Equivalent CPA would be classified as Low Risk of collision scenarios, regardless of the performance on every other criterion. The same reasoning is also true when considering the relationship between Equivalent CPA and a High Risk of collision classification.

The procedure used to define the veto threshold (v) is similar to the one used to define the indifference and preference thresholds. The thresholds for the two reference performance values (RP1 and RP2) are used to calculate the linear veto function parameters (α_v and β_v). The

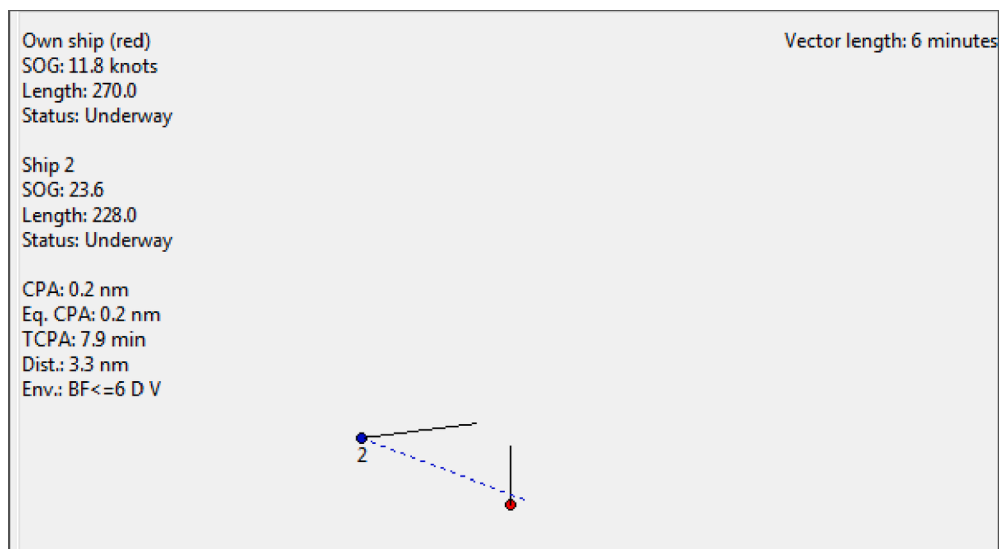
**Fig. 3.** Graphical representation of a validation scenario.

Table 7

Validation scenarios (Criteria performance table)

Scen. Nr.	Eq. CPA (nm)	TCPA (min.)	Dist. (nm)	Rel. Bear.	Env. Cond.	OS length (m)	OS status	TS length (m)	TS status
1	2.73	2.9	2.74	A	E8	112	Underway	218	Underway
2	1.13	32.2	2.27	C	E7	289	Restricted	238	Underway
3	2.14	0.7	2.16	B	E1	173	Underway	223	Underway
4	1.83	4.6	2.72	F	E2	166	Underway	203	Underway
5	0.35	25.1	4.98	B	E4	172	Underway	236	Underway
6	1.8	15.1	3.6	C	E2	195	Underway	229	Underway
7	0.66	39.1	5.97	B	E4	218	Underway	142	Underway
8	2.23	11.3	2.43	F	E6	40	Fishing	131	Underway
9	2.7	27.2	3.19	C	E3	140	Underway	97	Underway
10	0.62	13.3	1.97	E	E6	245	Underway	182	Underway
11	2.07	52	6.67	F	E6	222	Underway	171	Underway
12	0.75	19.5	2.08	F	E3	233	Underway	123	Underway
13	0.97	28.8	2.66	F	E7	158	Underway	187	Underway
14	2.8	38.1	2.93	B	E8	236	Underway	40	Fishing
15	0.17	21.7	2.44	F	E8	225	Underway	225	Underway
16	0.2	7.9	3.28	F	E1	270	Underway	228	Underway
17	1.55	32	5.23	A	E7	56	Underway	189	Restricted
18	1.84	29.7	3.15	B	E7	153	Underway	61	Fishing
19	1.01	22.5	2.72	F	E3	125	Underway	304	Underway
20	2.36	51.4	6.71	F	E6	265	Underway	99	Sailing
21	2.17	24.3	6.68	B	E5	93	Fishing	109	Underway
22	0.26	6.2	0.75	F	E7	175	Underway	54	Fishing
23	1.84	41.1	3.91	D	E7	300	Underway	289	Underway
24	2.17	42.8	6.45	E	E3	52	Underway	174	Underway
25	1.97	20.3	4.9	C	E2	180	Underway	186	Underway
26	0.77	10.4	3.15	B	E1	64	Underway	314	Underway
27	2.6	21.4	5.71	F	E3	144	Underway	227	Underway
28	2.52	27.9	2.69	B	E7	210	Underway	93	Underway
29	2.42	37.2	6.64	F	E2	250	Underway	89	Underway
30	2.99	29.6	7.52	B	E3	40	Underway	161	Underway

results are shown in Table 5.

5. Model validation

5.1. Validation results

To validate the model, a set of randomly generated scenarios were presented to the group of experts for classification. The scenarios were presented graphically, along with the performances of criteria not shown in the graphical representation. An example of a graphical representation of a scenario is shown in Figure 3. The criteria performances of the validation scenarios are shown in Table 6.

Validation was also performed by three individual experts not involved in the development of the model. The experience and qualifications of these external experts are presented in Table 6. The problem was contextualized as described in section 3. The experts were initially asked to classify the reference scenarios, to familiarize themselves with the process, and finally to classify the validation scenarios. Table 7 shows the collision risk categories assigned to each scenario by the expert panel that developed the model, by the three individual experts and by the model. A discrimination threshold $\lambda = 0.6$ was used for the ELECTRE Tri-nC method. The values of the credibility indices calculated using equations (9) and (10), and of the selecting function calculated using equation (11), are shown in Appendix 1, to detail the risk classification procedure.

5.2. Discussion

The validation results shown in Table 8 are very satisfactory. In 28 of the 30 validation scenarios (93.3%) the model assigned one possible category, which coincided with the expert panel's assignment. For one of the scenarios the model assigned two possible categories, one of them being the same as the experts' assignment. As for the remaining scenario, the group of experts hesitated between two categories, one of

which coincided with the model's assignment. Comparing the risk categories assigned by the model with the ones assigned by the external experts, it can be observed that the model assignments are compatible with the three experts' assignments for 21 of the scenarios (70%), with the assignments made by two of the experts in 7 cases (23.3%), and with the assignment made by only one of experts for 2 of the scenarios (6.7%). One of the scenarios was assigned to three different categories by the three experts.

As previously stated, a definition of risk was not provided to the experts. Rule 7 of the COLREGs does not mention ship types, the hazards associated with their cargoes or the number of human lives on board. During the sessions that led to the definition of criteria, there were some discussions on whether or not the ship type should be a criterion, but the final decision was not to include it. The reasoning provided was that a collision is undesirable regardless of the ship type. This indicates that the interpretation by the experts of the risk of collision as stated in rule 7 of the COLREGs is more closely related to the possibility of the collision occurring than to the eventual consequences of that event.

The results demonstrate that the model is suitable for assessing collision risk within the defined boundaries and limitations. Nonetheless, if a different definition of risk is discussed and agreed with the experts, additional criteria may be added. Also, this model can be combined within a probabilistic framework with other models that cover other risk components such as the static risk of the vessels involved in an encounter (e.g. age, flag, records of deficiencies and/or Port State Control detentions) as suggested by Dinis et al. [17], or the eventual consequences of a collision (e.g. type and quantity of cargo, quantity of fuel, number of crew and/or passengers).

6. Application - Assessment of collision avoidance options

In this section, the model is applied to compare different collision avoidance options for the encounter scenario shown in Figure 4. The model assigns a Medium Risk category to the initial state of the scenario.

Table 8

Validation results

Scenario Nr.	Exp. panel	Expert 1	Expert 2	Expert 3	Model
1	Low Risk	Low Risk	Low Risk	Low Risk	Low Risk
2	Medium Risk	Medium Risk	Medium Risk	LR/MR	Medium Risk
3	Low Risk	Low Risk	Low Risk	Low Risk	Low Risk
4	Low Risk	Low Risk	Low Risk	Low Risk	Low Risk
5	High Risk	High Risk	High Risk	MR/HR	High Risk
6	Low Risk	Medium Risk	LR/MR	Low Risk	Low Risk
7	Medium Risk	Medium Risk	High Risk	Low Risk	Medium Risk
8	Low Risk	Low Risk	Low Risk	Low Risk	Low Risk
9	Low Risk	Low Risk	Low Risk	Low Risk	Low Risk
10	High Risk	High Risk	High Risk	LR/MR	High Risk
11	Low Risk	Medium Risk	Low Risk	Low Risk	Low Risk
12	High Risk	High Risk	MR/HR	Medium Risk	MR/HR
13	Medium Risk	High Risk	MR/HR	Medium Risk	Medium Risk
14	Low Risk	Low Risk	LR/MR	Low Risk	Low Risk
15	High Risk	High Risk	High Risk	High Risk	High Risk
16	High Risk	High Risk	High Risk	High Risk	High Risk
17	Medium Risk	Medium Risk	Medium Risk	LR/MR	Medium Risk
18	Low Risk	Medium Risk	Low Risk	Low Risk	Low Risk
19	Medium Risk	MR/HR	LR/MR	Medium Risk	Medium Risk
20	Low Risk	Low Risk	LR/MR	Low Risk	Low Risk
21	Low Risk	Low Risk	Low Risk	Medium Risk	Low Risk
22	High Risk	High Risk	High Risk	High Risk	High Risk
23	LR/MR	Low Risk	Low Risk	Low Risk	Low Risk
24	Low Risk	Low Risk	Low Risk	Low Risk	Low Risk
25	Low Risk	Low Risk	Low Risk	LR/MR	Low Risk
26	Medium Risk	High Risk	High Risk	Medium Risk	Medium Risk
27	Low Risk	Low Risk	Low Risk	Low Risk	Low Risk
28	Low Risk	Low Risk	Low Risk	Low Risk	Low Risk
29	Low Risk	Low Risk	Medium Risk	LR/MR	Low Risk
30	Low Risk	Low Risk	Low Risk	Low Risk	Low Risk

Assuming no action is taken by any of the ships, the category assigned by the model changes to High Risk after 11 minutes ($t=11$), when the situation has evolved to the one shown in Figure 5. At this moment, it is assumed that the Own Ship (OS), depicted in red colour, alters its course to starboard (SB) to decrease the risk category. The impacts of six different course change options on the assigned risk category, as well as the category assignment when no course alteration is performed, are shown in Table 9, assuming the course alterations are performed instantly.

If no action is taken the assigned risk category will remain High Risk until the Closest Point of Approach is reached, at $t=22.5$ minutes. A course alteration of 10° to starboard results in a new risk category assignment of Medium Risk/High Risk, which increases again to High Risk after one minute, at $t=12$ minutes. A course alteration of 20° to starboard lowers the risk category to medium risk, but an increase to High Risk will occur at $t=19$ minutes. Course alterations of 30° , 40° or 50° to starboard (Figure 6) ensure a risk reduction to Medium Risk that remains at this level until the Closest Point of Approach is passed and the distance between ships starts increasing. A course alteration of 60° to starboard (Figure 7) decreases the risk category to Low Risk until approximately the time of Closest Point of Approach, when the risk category assigned by the model becomes Low Risk/Medium Risk for brief moments before the distance between ships starts increasing and the situation is resolved.

7. Conclusion

The ELECTRE Tri-nC expert-based decision aiding method is used to successfully develop and implement a model that assigns ship encounter scenarios to three risk categories. The model considers criteria related to the ships' motion, but also to ship characteristics and environmental factors. It is concluded that the method is adequate for modelling the preferences of experts in a collision risk assessment context, although it should be noted that the application was limited to near ideal scenarios involving only two vessels in open waters.

The model is validated by comparing its assignments of a set of randomly generated scenarios to the risk categories with the assignments made by the experts, for the same set of scenarios. For 93.3% of the scenarios the assignment made by both the model and the experts is the same risk category. For the remaining two scenarios, either the risk

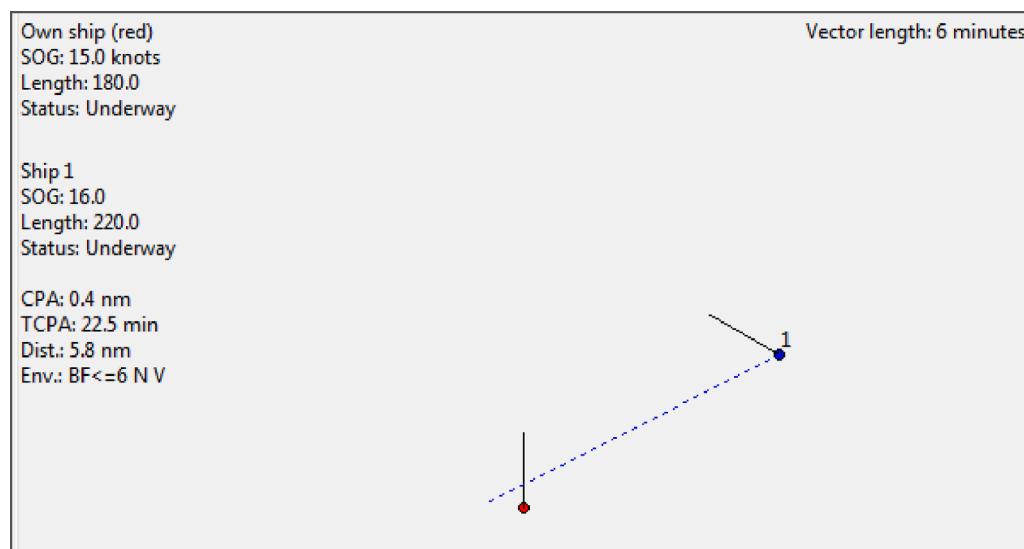


Fig. 4. Application scenario, initial condition ($t=0$ minutes).

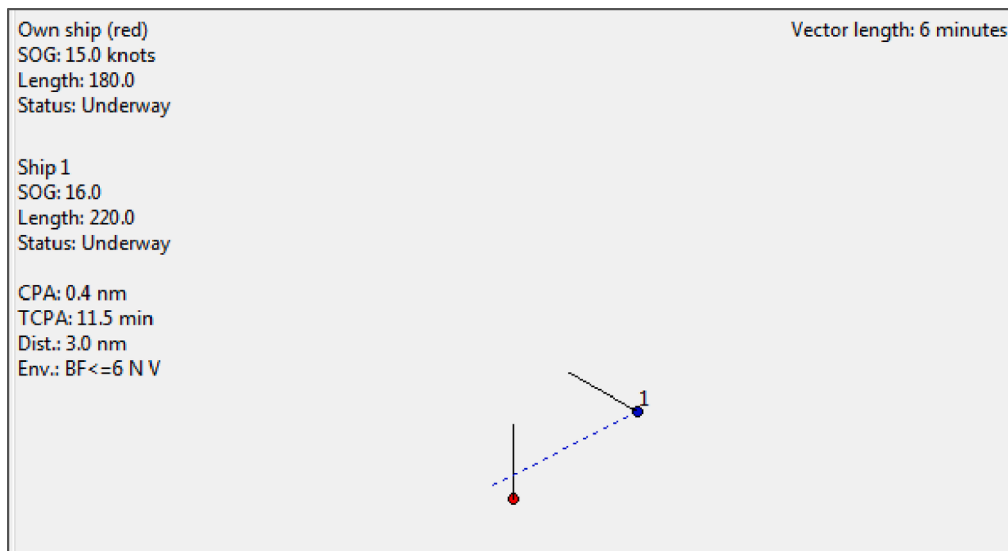


Fig. 5. Application scenario, category increased to High Risk (t=11 minutes).

Table 9

Effect of course alterations to starboard on assigned risk category.

	Course alteration to starboard (SB), performed at t=11 minutes						
	0°	10° SB	20° SB	30° SB	40° SB	50° SB	60° SB
Assigned risk category at t=11 minutes (after course alteration)	High Risk	MR/HR	Medium Risk	Medium Risk	Medium Risk	Medium Risk	Low Risk
Subsequent risk category change (at t [minutes])	No change	HR (t=12)	HR (t=19)	No change	No change	No change	LR/MR (t=16)

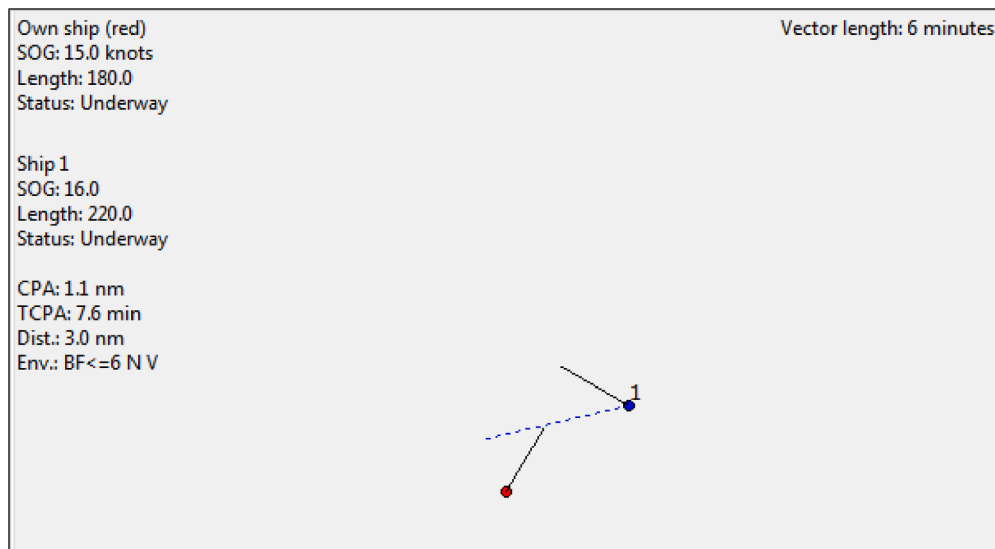


Fig. 6. Application scenario, course alteration of 30° SB at t=11 minutes.

category assigned by the model is one of two assigned by the experts, or vice-versa. These results indicate that the model accurately reflects the experts' opinion regarding the assessment of ship collision risk.

A subsequent validation procedure is performed, involving three experts that have not participated in the development of the model. Although a limited number of experts participated in this validation process, the results are also satisfactory, since the model's assignments are compatible with the assignments made by the three experts for 70% of the scenarios and with the assignments made by two of the experts in 23.3% of the cases.

The model is applied to an encounter scenario between two vessels to

monitor collision risk as the situation evolves and to assess the effect of possible course alterations on the assigned risk category, both at the moment of the collision avoidance action and subsequently, until the encounter situation is resolved and the distance between ships starts increasing. This application demonstrates the possible use of the model as an aid for real-time collision risk assessment and collision avoidance, both on board ships and at VTS centres.

Possible future developments include expanding the model to classify the risk of collision when more than two ships are involved, as well as testing the robustness to expert subjectivity by validating the results using an enlarged and different set of experts that have not participated

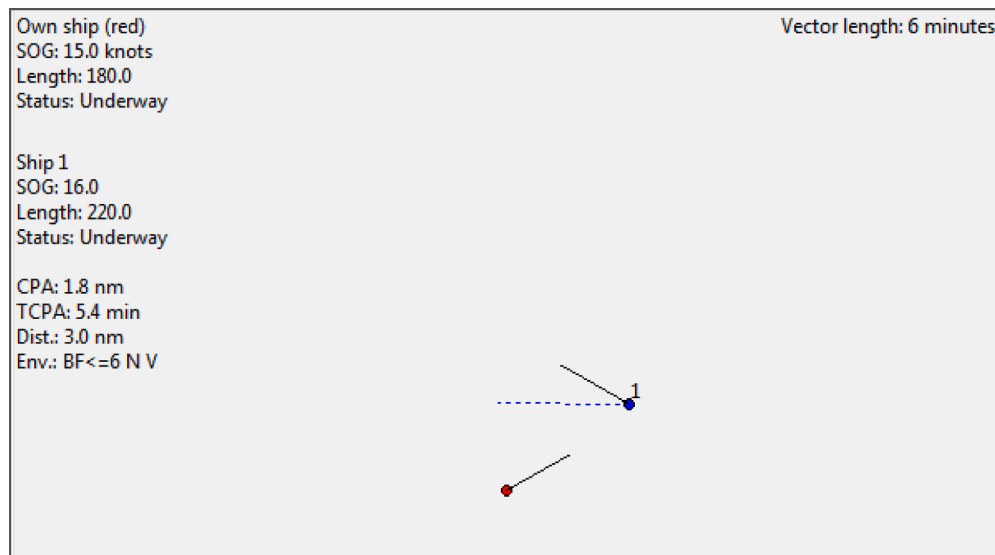


Fig. 7. Application scenario, course alteration of 60° SB at t=11 minutes.

Table 10

Credibility of each validation scenario outranking each risk category and vice-versa.

Scenario Nr.	$\sigma(\{a\}, LR)$	$\sigma(\{a\}, MR)$	$\sigma(\{a\}, HR)$	$\sigma(LR, \{a\})$	$\sigma(MR, \{a\})$	$\sigma(HR, \{a\})$
1	1.0	0.0	0.0	0.470	0.686	1.0
2	1.0	0.774	0.0	0.0	0.707	1.0
3	1.0	0.0	0.0	0.418	0.774	1.0
4	1.0	0.0	0.0	0.419	0.636	1.0
5	1.0	0.740	0.271	0.0	0.0	1.0
6	1.0	0.0	0.0	0.417	0.849	1.0
7	1.0	0.570	0.312	0.0	0.904	1.0
8	1.0	0.0	0.0	0.282	0.644	1.0
9	1.0	0.0	0.0	0.542	0.853	1.0
10	1.0	1.0	0.774	0.0	0.541	1.0
11	1.0	0.0	0.0	0.735	0.904	1.0
12	1.0	0.908	0.676	0.0	0.730	1.0
13	1.0	0.955	0.0	0.0	0.756	1.0
14	1.0	0.0	0.0	0.633	0.893	0.989
15	1.0	1.0	0.774	0.0	0.0	1.0
16	1.0	0.904	0.678	0.0	0.0	1.0
17	1.0	0.466	0.0	0.0	0.904	1.0
18	1.0	0.0	0.0	0.316	0.893	0.989
19	1.0	0.859	0.0	0.0	0.844	1.0
20	1.0	0.0	0.0	0.717	0.904	1.0
21	1.0	0.0	0.0	0.497	0.859	1.0
22	1.0	0.954	0.948	0.0	0.0	0.989
23	1.0	0.0	0.0	0.500	0.896	1.0
24	1.0	0.0	0.0	0.601	0.853	1.0
25	1.0	0.0	0.0	0.635	0.853	1.0
26	1.0	0.859	0.408	0.0	0.885	1.0
27	1.0	0.0	0.0	0.536	0.997	1.0
28	1.0	0.0	0.0	0.429	0.904	1.0
29	1.0	0.0	0.0	0.638	1.0	1.0
30	1.0	0.0	0.0	0.689	1.0	1.0

in the development of the model.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 11

Values of selecting function, risk category assigned to each validation scenario.

Scenario Nr.	$\rho(\{a\}, LR)$	$\rho(\{a\}, MR)$	$\rho(\{a\}, HR)$	Model assignment
1	0.470	0.0	0.0	Low Risk
2	0.0	0.707	0.0	Medium Risk
3	0.418	0.0	0.0	Low Risk
4	0.419	0.0	0.0	Low Risk
5	0.0	0.0	0.271	High Risk
6	0.417	0.0	0.0	Low Risk
7	0.0	0.570	0.312	Medium Risk
8	0.282	0.0	0.0	Low Risk
9	0.542	0.0	0.0	Low Risk
10	0.0	0.541	0.774	High Risk
11	0.735	0.0	0.0	Low Risk
12	0.0	0.730	0.676	MR/HR
13	0.0	0.756	0.0	Medium Risk
14	0.633	0.0	0.0	Low Risk
15	0.0	0.0	0.774	High Risk
16	0.0	0.0	0.678	High Risk
17	0.0	0.466	0.0	Medium Risk
18	0.316	0.0	0.0	Low Risk
19	0.0	0.844	0.0	Medium Risk
20	0.717	0.0	0.0	Low Risk
21	0.497	0.0	0.0	Low Risk
22	0.0	0.0	0.948	High Risk
23	0.500	0.0	0.0	Low Risk
24	0.601	0.0	0.0	Low Risk
25	0.635	0.0	0.0	Low Risk
26	0.0	0.859	0.408	Medium Risk
27	0.536	0.0	0.0	Low Risk
28	0.429	0.0	0.0	Low Risk
29	0.638	0.0	0.0	Low Risk
30	0.689	0.0	0.0	Low Risk

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taking the time to share their knowledge and experience.

Appendix 1. Values of credibility indices and of rho function

The values of the credibility indices calculated using equations (9) and (10) are shown in Table 10. The values of the selecting function ρ calculated using equation (11) and the categories assigned by the model to each scenario are shown in Table 11.

Taking as an example scenario 7, both the descending and the ascending rules described in section 3.2 assign the same category (Medium Risk), so this is the risk category assigned by the method to that scenario.

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